

12.2 Centripetal Acceleration

Question Paper

Course	CIE A Level Physics
Section	12. Motion in a Circle
Topic	12.2 Centripetal Acceleration
Difficulty	Medium

Time allowed: 30

Score: /20

Percentage: /100

Question 1a

Two runners are running around a circular track. One runner follows path X and the other follows path Y, as shown in Fig. 1.1.

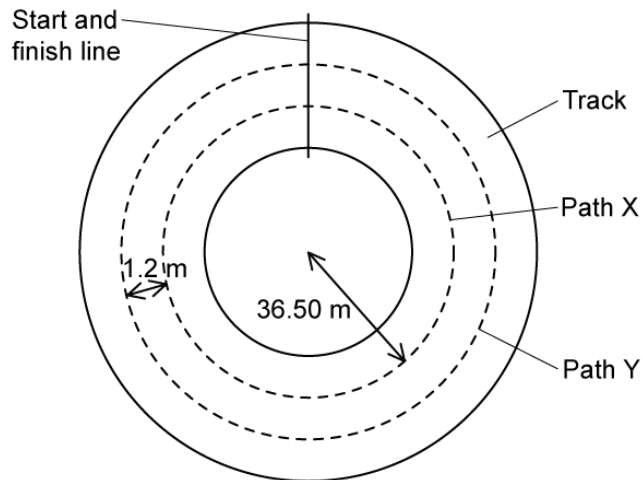


Fig 1.1

The radius of path X is 36.50 m. Path Y is parallel to, and 1.2 m outside, path X. Both runners have mass 75 kg. The maximum lateral (sideways) friction force F that the runners can experience without sliding is 57 mN.

Show that the maximum speed at which the runner on path Y can move is 0.17 m s^{-1} .

[2 marks]

Question 1b

Compare the centripetal acceleration and maximum speed of the runner on path X compared to path Y.

[2 marks]

Question 1c

Calculate the time taken for the runner in path X to complete 1 lap.

$T = \dots\dots\dots$ minutes

[2 marks]

Question 1d

The runner on path Y has an ankle injury that has come back with a vengeance, so it takes them twice as long to complete 1 lap.

Determine their new centripetal acceleration.

[3 marks]

Question 2a

A bowl-shaped skating ramp can be modelled as a hollow sphere.

A skateboarder with a horizontal speed follows a circular path inside the bowl of radius 2.3 m, as shown in Fig 1.1.

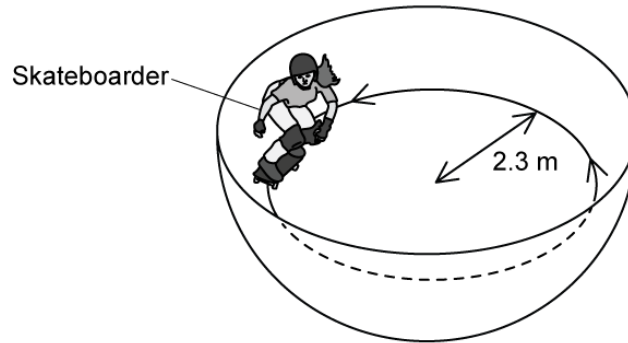


Fig 1.1

The forces acting on the skateboarder are their weight W and the normal reaction force R of the ramp on the skateboarder. R is at an angle θ to the horizontal.

Sketch W , R and θ on Fig 1.2.

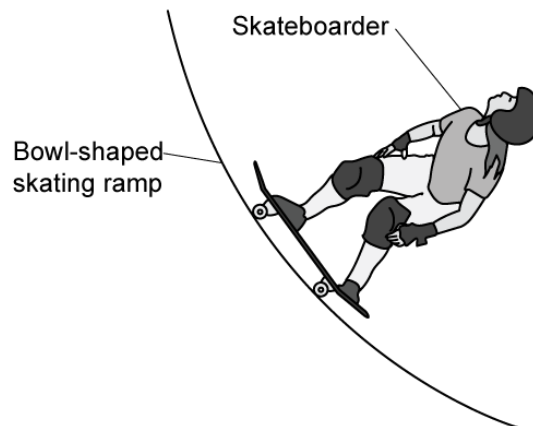


Fig 1.2

[3 marks]

Question 2b

Determine an equation for W in terms of θ and the resultant force, F on the skateboarder.

[3 marks]**Question 2c**

Explain why force F is significant to the motion of the skateboarder.

[2 marks]**Question 2d**

For the radius of the ramp in Fig 1.1, the skateboarder travels at a speed of 6.0 m s^{-1} .

Show that the angle θ is 32° .

[3 marks]

Model Answers: Medium

1a

(a) Show that the maximum speed at which the runners can move around the track without sliding is 0.17 m s^{-1} .

List the known quantities:

- Mass of each runner, $m = 75 \text{ kg}$
- Radius of circular motion, $r = 36.50 + 1.2 = 37.7 \text{ m}$
- Frictional (centripetal) force, $F = 57 \text{ mN} = 57 \times 10^{-3} \text{ N}$

Determine an equation for the maximum speed, v :

- $F = \frac{mv^2}{r} \Rightarrow v = \sqrt{\frac{Fr}{m}}$
- $v = \sqrt{\frac{(57 \times 10^{-3}) \times 37.7}{75}}$ [1 mark]
- $v = 0.169 = 0.17 \text{ m s}^{-1}$ [1 mark]

[Total: 2 marks]

1b

(b) The comparison between the centripetal acceleration and maximum speed of the runner on path X compared to path Y is:

- They both have the same centripetal acceleration; [1 mark]
- The runner on path X has a smaller maximum speed (than the runner on path Y); [1 mark]

[Total: 2 marks]

1c

(c) Calculate the time taken for the runner in path X to complete 1 lap

List the known quantities:

- Mass of each runner, $m = 75 \text{ kg}$
- Radius of circular motion, $r = 36.50 \text{ m}$
- Frictional (centripetal) force, $F = 57 \text{ mN} = 57 \times 10^{-3} \text{ m}$

Determine an equation for the time period, T :

- $F = m\omega^2 r = m \left(\frac{2\pi}{T} \right)^2 r$
- $T = \sqrt{\frac{m4\pi^2 r}{F}}$

Substitute the values into the time period equation:

- $T = \sqrt{\frac{75 \times 4\pi^2 \times 36.50}{57 \times 10^{-3}}} \text{ [1 mark]}$
- $T = 1376.95 \text{ s}$

Convert the time period into minutes:

- $T = \frac{1376.95}{60} = 22.95 = 23 \text{ minutes [1 mark]}$

[Total: 2 marks]

Make sure you read the question carefully! This is now talking about the runner on path X, so we are now using a different value of r (but both runners still have the same F).

Always look out for the units required in the final answer. Although time period will normally be given in seconds, we must convert it into **minutes** here.

1d

(d) Determine their new centripetal acceleration

List the known quantities:

- Mass, $m = 75 \text{ kg}$
- Previous frictional (centripetal) force, $F = 57 \text{ mN} = 57 \times 10^{-3} \text{ m}$

Calculate the previous centripetal acceleration, a_p :

- $F = ma \Rightarrow a = \frac{F}{m}$
- $a_p = \frac{57 \times 10^{-3}}{75} = 7.6 \times 10^{-4} \text{ m s}^{-2}$ [1 mark]

Determine the change in centripetal acceleration:

- $a = r\omega^2 = r\left(\frac{2\pi}{T}\right)^2 = \frac{r4\pi^2}{T^2}$
- $a \propto \frac{1}{T^2}$
- Therefore, if T doubles, then a decreases by 4

Calculate the new centripetal acceleration, a :

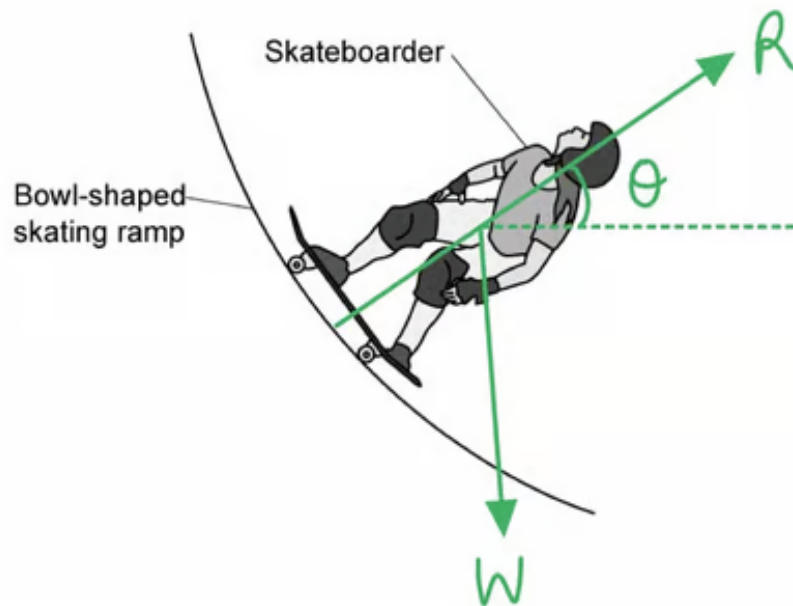
- $a = \frac{a_p}{4} = \frac{7.6 \times 10^{-4}}{4}$ [1 mark]
- $a = 1.9 \times 10^{-4} \text{ m s}^{-2}$ [1 mark]

[Total: 3 marks]

2a

(a) Sketch W , R and θ on Fig 1.2

A sketch that would score 3 marks should look like:



- R from the skating ramp directed normal to the skating ramp; [1 mark]
- W directly downwards from the centre of mass of skateboarder; [1 mark]
- θ labelled between a horizontal line and R ; [1 mark]

[Total: 3 marks]

R is the reaction force from the skating ramp on the skateboarder. Therefore, it must start **on the ramp** and not at the centre of the mass of the skateboarder as the weight does! If the arrows are from an incorrect point, you will not get all the marks.

2b

(b) Determine an equation for W in terms of θ and the resultant force, F on the skateboarder

Equate the components of R to F and W :

- $F = R \cos \theta$ [1 mark]
- $W = R \sin \theta$ [1 mark]

Determine an equation in terms of W , θ and F :

- $$\frac{W}{F} = \frac{R}{R} \frac{\sin \theta}{\cos \theta} = \tan \theta$$

[Total: 3 marks]

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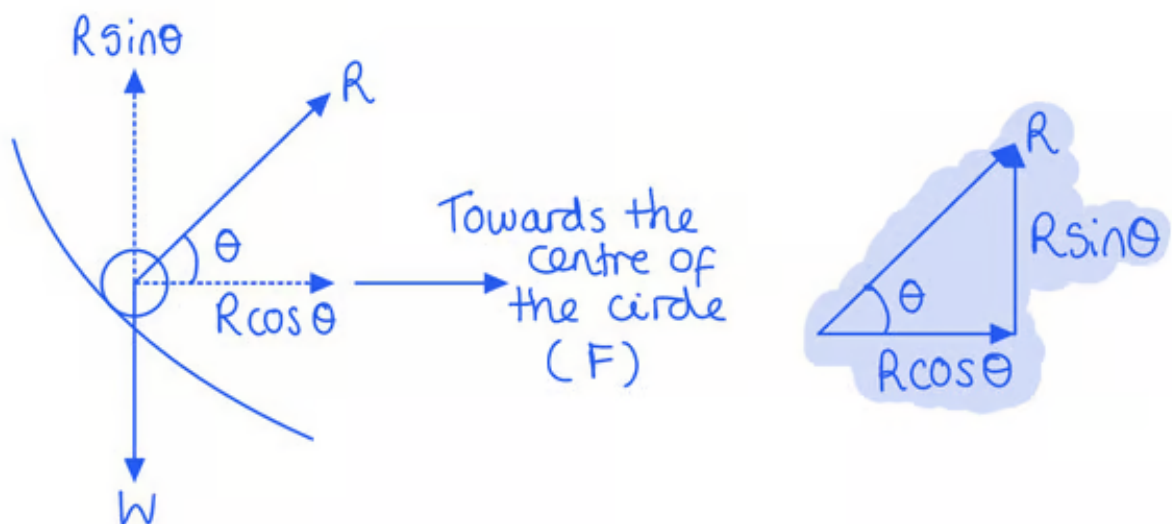
- $F = R \cos \theta$ [1 mark]
- $W = R \sin \theta$ [1 mark]

Determine an equation in terms of W , θ and F :

- $\frac{W}{F} = \frac{R \sin \theta}{R \cos \theta} = \tan \theta$
- $W = F \tan \theta$ [1 mark]

[Total: 3 marks]

The horizontal component, $R \cos \theta$ is in the same direction as the resultant force, F (towards the centre of the circle). The vertical component $R \sin \theta$ is equal to the weight W , because these forces are balanced (there is no resultant force in the vertical direction). This might be easier to relate to with a sketch:



2c

(c) Force F is significant to the motion of the skateboarder because:

- It provides the centripetal force; [1 mark]
- To keep the skateboarder moving in a circle; [1 mark]

[Total: 2 marks]

2d

(d) Show that the angle θ is 32°

List the known quantities:

- Radius of the path, $r = 2.3 \text{ m}$
- Speed of the skateboarder, $v = 6.0 \text{ m s}^{-1}$
- Acceleration of free fall, $g = 9.81 \text{ m s}^{-2}$

Determine an equation for the angle, θ :

- $W = F \tan \theta$ (from part b)
- $W = mg$ and $F = \frac{mv^2}{r}$
- $mg = \frac{mv^2}{r} \tan \theta$
- $\tan \theta = \frac{rg}{v^2} \Rightarrow \theta = \tan^{-1} \left(\frac{rg}{v^2} \right)$ [1 mark]

Calculate the angle, θ :

- $\theta = \tan^{-1} \left(\frac{rg}{v^2} \right) = \tan^{-1} \left(\frac{2.3 \times 9.81}{6.0^2} \right)$ [1 mark]
- $\theta = 32^\circ$ [1 mark]

[Total: 3 marks]